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Minimum-Error Quantum State Discrimination Based on Semidefinite Programming*

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Abstract: In this paper we consider the *general* problem of designing the optimal quantum detection for *minimum-error quantum state discrimination*. Since it is hard to get an explicit solution and we must resort to *numerical* methods for the general case. We show the problem can be cast into a semidefinite programming(SDP) problem, we use the theory of SDP to solve it. Based on the idea, we propose a method for minimum-error discrimination by formulating the design of optimal quantum detection as a standard SDP problem; furthermore, based on the optimality conditions for SDP, we derive a set of optimality conditions for quantum detection. It is exemplified that this method is simple to implement and can be used to design optimum detection operators for minimum-error discrimination efficiently.

Key Words: Semidefinite programming, Quantum states, Minimum-error discrimination

1 INTRODUCTION

Quantum information and quantum communication [1]is now a well-developed field of research, and classical information is transmitted by using quantum states and channels. The carrier of information is some quantum system and information is encode in its state. The transmitter (Alice) encodes a message onto a set of quantum states. Based on the knowledge of the quantum states and their prior probabilities, the task of the receiver (Bob) is to decode the received message by finding optimal measurement strategy. However, quantum mechanically, non-orthogonal quantum states can't be reliably distinguished, and, thus, a fundamental problem arises: after the desired transformation is performed on the input state by the quantum process, the state of the system has to be determined. Therefore, two different strategies, optimum unambiguous discrimination[2-5] and minimum-error discrimination[6-11], have been developed to distinguish quantum states. Optimum unambiguous discrimination allows an inconclusive result that fails to give a definite answer after the distinguishing measurement. When the probability of failure is minimum, optimum unambiguous discrimination is realized. Another minimum-error discrimination, yielding a conclusive outcome which may be wrong after the distinguishing measurement, is to minimize the probability of detection error. In this paper, we will consider the strategy for minimum-error quantum state discrimination.

Minimum-error discrimination as well as optimum unambiguous discrimination have been extensively studied over past few years. However, the explicit analytical form for optimal measurement strategy has been found only for some special cases. These include the cases of only two quantum states [6], symmetric states [10, 11], mirror-symmetric states [9], linearly independent states [12], and equiprobable states that are complete in the sense that a weighted sum of projectors onto the states equals the identity operator [7]. For the general case, it is still hard to get the analytical expressions of optimal detection operators. Therefore, different numerical methods have been proposed [13-15]. Theose methods mainly include iteration algorithms involving Lagrange multipliers [14] and SDP method [15]. However, in our opinion, although existing iterative algorithms have been proved to be feasible, the main disadvantages of them are that they converge slowly and the solution obtained is not guaranteed to converge towards the global optimum. As to the SDP method, we deem it has several advantages. On one hand, a SDP is a convex optimization problem, and correspondingly has the very desirable property that a local minimum is automatically a global minimum. On the other hand, many efficient algorithms for solving SDPs are developed, by which the SDPs can be computed in polynomial time. However, the existing methods are not convenient to implement since these constructed SDP problems have very complicated structure.

In this paper we demonstrate how the *general problem* of minimum-error discrimination can be formulated as a standard SDP problem. After that, we propose a SDP based numerical method for the general problem. Furthermore, based on the optimality conditions for SDP, we derive a set of optimality conditions for detection operators. In addition, we obtain optimal value of the correct detection probability and optimal detection operators numerically using the software $SeDuMi$ which solves the proposed SDP problems efficiently.

The paper is organized as follows. In section 2 we will briefly introduce the theory of SDP which lay the foundation of this paper. In section 3 we will propose our numerical method based on the theory of SDP. In section 4 we will derive the optimality conditions for optimal detection operators. Finally in section 5, we illustrate the SDP method through a worked out example.

2 PRELIMINARIES

SDP [16] is a problem of minimizing a linear function subject to the constraint that an affine combination of symmetric matrices is positive semidefinite. Such a constraint is nonlinear and nonsmooth, but convex, so semidefinite programs are

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convex optimization problems.
A standard SDP is the problem of minimizing a linear function of a variable $x \in R^M$ subject to a matrix inequality:

$$\begin{aligned} PSDP \qquad & \min \quad c^{\mathrm{T}}x \\ subject\ to \quad & F(x) \geqslant O \\ where \quad & F(x) = F_0 + \sum_{i=1}^{M} x_i F_i \end{aligned} \tag{1}$$

The problem data are the vector $c \in R^M$ and symmetric matrices $F_0, \ldots, F_M \in R^{N\times N}$. The inequality sign in $F(x) \geqslant 0$ means $F(x)$ is positive semidefinite.
The problem of (1) is referred to as *the primal problem*. x is said to be *primal feasible* if $F(x) \geqslant 0$, and *strictly primal feasible* if $F(x) > 0$. When there exists a *strictly feasible point*, the $PSDP$ is said to be *strictly feasible*.
The *dual problem* associated with (1) is

$$\begin{aligned} DSDP \qquad & \max \quad -Tr(F_0 Z) \\ subject\ to \quad & Tr(F_i Z) = c_i, i = 1, \ldots, M \\ & Z \geqslant 0 \end{aligned} \tag{2}$$

Here the variable is the matrix $Z = Z^{\mathrm{T}} \in R^{N\times N}$, which is subject to M equality constrains and the matrix nonnegativity condition.
We will say that $Z = Z^{\mathrm{T}} \in R^{N\times N}$ is *dual feasible* if $Tr(F_i Z) = c_i$, $i = 1, \ldots, M$, $Z \geqslant 0$, and is *strictly dual feasible* if $Tr(F_i Z) = c_i$, $i = 1, \ldots, M$, $Z > 0$. If there exists a *strictly feasible point*, then the dual problem is said to be *strictly feasible*.
It is shown [16] that the optimality conditions for SDP are as follows which justify the name *dual*:
Assume that both $PSDP$ and $DSDP$ are strictly feasible, we say x (or Z) is optimal if and only if there is a Z (or x) such that

$$\begin{aligned} F(x) &\geqslant O \\ Z \geqslant 0, Tr(F_i Z) &= c_i, i = 1, \ldots, M \\ F(x)Z &= ZF(x) = O \end{aligned} \tag{3}$$

Based on above theory of SDP, later we will discuss SDP formulation of minimum-error discrimination and optimality conditions for minimum-error discrimination in the following sections.

3 STANDARD SDP FORMULATION OF MINIMUM-ERROR DISCRIMINATION

3.1 General Problem of Minimum-error Discrimination

Suppose that Alice selects one state at random to send to Bob from a given set of quantum states $\{\rho_j\}(j = 1, 2, \ldots, m)$, and tells him the set of states and the prior probability ξ_j for each of the states $\rho_j$where $\sum_{j=1}^{m} \xi_j = 1$, then Bob's task is to devise m optimal detection operators Π_j satisfying $O \leqslant \Pi_j \leqslant I$ and $\sum_{j=1}^{m} \Pi_j = I$, to minimize the error probability or equivalently, maximize the correct probability of detecting the state. Each of the detection operators Π_j corresponds to one measurement outcome, labeled by j . Then the probability that Bob obtains the outcome k while the true conveyed state is j is $P(k|j) = Tr(\Pi_k \rho_j)$. Therefore, the average probability of correct detection is given by

$$P_s = \sum_{j=1}^{m} \xi_j Tr \Pi_j \rho_j \tag{4}$$

Since the objective is to keep $Bob's$ error rate as low as possible, the detection problem reads

$$\begin{aligned} \max \quad & \sum_{j=1}^{m} \xi_j Tr \Pi_j \rho_j \\ subject\ to \quad & \sum_{j=1}^{m} \Pi_j = I \\ & \Pi_j \geqslant O, j = 1, \ldots, m \end{aligned} \tag{5}$$

where $\rho_j, \Pi_j (j = 1, \ldots, m)$ are $n \times n$ matrices.
Denoting $\rho_j' = \xi_j \rho_j$, we write problem (5) as

$$\begin{aligned} \max \ & -Tr \sum_{j=1}^{m} \Pi_j (-\rho_j') \\ subject\ to \quad & \sum_{j=1}^{m} \Pi_j = I \\ & \Pi_j \geqslant O, j = 1, \ldots, m \end{aligned} \tag{6}$$

3.2 Standard SDP Formulation of Minimum-error Discrimination

By comparing (6) with (2) , we find they are similar in the appearance. It is obvious that (6) is a SDP problem, thus we can transform (6) into a standard SDP formulation. The standard SDP formulation of minimum-error discrimination is given by

Theorem 1 The general problem of minimum-error discrimination can be transformed into such a standard SDP whose parameters are

$$\begin{aligned} F_0 &= \mathrm{diag}(-\rho_1', \ldots, -\rho_m'), c_i = \delta_{lk} \\ F_i &= \frac{1}{2} \sum_{j=1}^{m} (|e_{l+(j-1)n} >< e_{k+(j-1)n}|) + \\ & \quad \frac{1}{2} \sum_{j=1}^{m} (|e_{k+(j-1)n} >< e_{l+(j-1)n}|) \\ & \quad i = l + (k-1)n, l, k = 1, \ldots, n \end{aligned} \tag{7}$$

Here $\delta_{lk} = \begin{cases} 1 & l = k \\ 0 & l \neq k \end{cases}$ denotes the $Kronecker's$ delta, and $|e_i > (i = 1, \ldots, mn)$ is a set of basis vectors.
Proof To formulate (6) as a standard SDP , we first construct

$$Z = Z^{\mathrm{T}} = \mathrm{diag}(\Pi_1, \ldots, \Pi_m) \tag{8}$$

$$F_0 = \mathrm{diag}(-\rho_1', -\rho_2' \ldots, -\rho_m') \tag{9}$$

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Obviously,

$$-TrF_0Z = -Tr\sum_{j=1}^{m}\Pi_j(-\rho_j') = \sum_{j=1}^{m}\xi_j Tr\Pi_j\rho_j \quad (10)$$

Rewritten in terms of the matrix elements$< e_l|A|e_k >= a_{lk}$, $\sum_{j=1}^{m}\Pi_j = I$ reads

$$\begin{aligned}\delta_{lk} &=< e_l|\sum_{j=1}^{m}\Pi_j|e_k >= \sum_{j=1}^{m} < e_l|\Pi_j|e_k > \\ &= \sum_{j=1}^{m} < e_{l+(j-1)n}|Z|e_{k+(j-1)n} > \\ &\qquad (l,k=1,\ldots,n)\end{aligned} \quad (11)$$

i.e.

$$Tr(\sum_{j=1}^{m}|e_{k+(j-1)n}>< e_{l+(j-1)n}|Z) = \delta_{lk} \quad (12)$$

In (12), denoting

$$F_{lk} = \sum_{j=1}^{m}|e_{k+(j-1)n}>< e_{l+(j-1)n}|(l,k=1,\ldots,n) \quad (13)$$

we express (12) as

$$Tr(F_{lk}Z) = \delta_{lk},\ \ l,k=1,\ldots,n \quad (14)$$

Note that the matrix F_{lk} in (13) is not symmetric. According to the theory of SDP, we need construct a symmetric matrix F^{lk} such that $Tr(F^{lk}Z) = c^{lk}$.
Now we define F^{lk} and c^{lk} by

$$F^{lk} = \frac{F_{kl}+F_{lk}}{2}, c^{lk} = \delta_{lk},\ \ l,k=1,\ldots,n \quad (15)$$

It is easy to show that (15) verifies $Tr(F^{kl}Z) = c^{lk}$ and $F^{lk} = (F^{lk})^{\mathrm{T}}$.
Using the single index i to denote the double (l,k) with $i = l+(k-1)n$,we can write (14) as

$$Tr(F_iZ) = c_i \quad (16)$$

Since $\Pi_j \geqslant O,\ j=1,\ldots,m$,this implies

$$Z = diag(\Pi_1,\ldots,\Pi_m) \geqslant O \quad (17)$$

Thus we see (6) can be formulated as a standard SDP formulation and the parameters of the SDP are given by (7). This completes the proof.
Based on theorem 1, we obtain a numerical method of minimum-error quantum state discrimination. By solving the standard SDP whose parameters are given by (7), we can get the optimal solution of the SDP. Then, according to the one-one relation between Z and Π_j, we can design the optimal detection operators for minimum-error discrimination.

4 THE OPTIMALITY CONDITIONS FOR DETECTION OPERATORS

Based on the theorem 1 and the optimality conditions for SDP, we will derive the optimality conditions for detection operators in this section.
It is easy to verify that both the $PSDP$ and the $DSDP$ of minimum-error discrimination are strictly feasible. Therefore we can directly apply the optimality conditions for SDP to derive the optimality conditions for detection operators which are as follows
Theorem 2 $\{\Pi_j(j=1,\ldots,m)\}$ is the optimal detection operators if and only if there is a matrix $X \in R^{n\times n}$ such that

$$\begin{cases} \begin{bmatrix} X-\rho_1' & & \\ & \ddots & \\ & & X-\rho_m' \end{bmatrix} \geqslant O \\ \begin{bmatrix} X-\rho_1' & & \\ & \ddots & \\ & & X-\rho_m' \end{bmatrix} Z = O \\ Z\begin{bmatrix} X-\rho_1' & & \\ & \ddots & \\ & & X-\rho_m' \end{bmatrix} = O \end{cases} \quad (18)$$

Since it is not hard to demonstrate theorem 2 [16], we will not give the proof here.
Furthermore , (18) can be simplified into

$$\begin{aligned} X-\rho_i' &\geqslant O, i=1,\ldots,m \\ (X-\rho_i')\Pi_i &= \Pi_i(X-\rho_i') = O, i=1,\ldots,m \end{aligned} \quad (19)$$

In second set of equations of (19), sum over i and using$\sum_{i=1}^{m}\Pi_i = I$, we obtain

$$X = \sum_{i=1}^{m}\Pi_i\rho_i' = \sum_{i=1}^{m}\rho_i'\Pi_i \quad (20)$$

Substitute (20) into (19), we get

$$\begin{cases} \sum_{j=1}^{m}\Pi_j\rho_j' \geqslant \rho_i', i=1,\ldots,m \\ \sum_{i=1}^{m}\Pi_i\rho_i' = \sum_{i=1}^{m}\rho_i'\Pi_i \end{cases} \quad (21)$$

Therefore, (21) is also the optimality conditions for detection operators, which is identical to the results of Yuen, *et al* [7]

5 EXAMPLE

To illustrate our numerical method for minimum-error discrimination, we will consider a specific example [11].
Consider the problem in which the set of quantum states consists of three pure states described by density operators

$$\rho_i = |\phi_i>< \phi_i|, i=1,2,3$$

where
$|\phi_1>= \begin{bmatrix} 1 \\ 0 \end{bmatrix}, |\phi_2>= \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}, |\phi_3>= \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

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with prior probabilities

$$\xi_1 = 0.1, \xi_2 = 0.6, \xi_3 = 0.3$$

We will seek the optimal detection operators $\Pi_1, \Pi_2, and\, \Pi_3$ to minimize the error probability of detection.
The computational steps are as follows
First, by (5), the problem reads

$$\begin{aligned} & \max \quad \sum_{j=1}^{3} Tr\Pi_j \rho_j' \\ subject\ to \quad & \Pi_1 + \Pi_2 + \Pi_3 = I \\ & \Pi_j \geqslant 0, j = 1, 2, 3 \end{aligned} \tag{22}$$

where $\rho_j' = \xi_j \rho_j \quad (j = 1, 2, 3).$

Second, according to theory of SDP and theorem 1, we can formulate problem (22) as a standard SDP which is the special case of (1) (M=3). By (8), we can get $Z = \mathrm{diag}(\Pi_1, \Pi_2, \Pi_3)$, and the parameters of SDP are given by (7).
Next, based on the standard SDP formulation of problem (22), we can directly compute the optimal value of the detection probability and the optimal detection operators by using the software package $SeDuMi1.05$ in $Matlab7.0$. From the output, we have
(1) iteration numbers: 9
(2) iteration time: 0.6 s
(3) the optimal value of the correct detection probability:
$\max \quad \sum_{j=1}^{3} Tr\Pi_j \rho_j' = 7.8541019652e^{-}001$
(4) the optimal solution of $PSDP$ of (22):
$x = (0.3512, 0.4342, 0, 0.4342)'$
(5) the optimal solution of $DSDP$ of (22)

$$Z = \begin{bmatrix} 0 & 0 & -0.0000 & -0.0000 & -0.0000 & 0.0000 \\ 0 & 0 & 0.0000 & 0.0000 & 0.0000 & -0.0000 \\ 0 & 0 & 0.7236 & 0.4472 & -0.0000 & 0.0000 \\ 0 & 0 & 0.4472 & 0.2764 & -0.0000 & 0.0000 \\ 0 & 0 & -0.0000 & -0.0000 & 0.2764 & -0.4472 \\ 0 & 0 & 0.0000 & 0.0000 & -0.4472 & 0.7236 \end{bmatrix} \tag{23}$$

By the one-one relation of Z and $\Pi_i (i = 1, \cdots, m)$ in (8), we directly get the optimal detection operators

$$\Pi_1 = O$$

$$\Pi_2 = \begin{bmatrix} 0.7236 & 0.4472 \\ 0.4472 & 0.2764 \end{bmatrix}$$

$$\Pi_3 = \begin{bmatrix} 0.2764 & -0.4472 \\ -0.4472 & 0.7236 \end{bmatrix}$$

Finally, we can verify that detection operators (24) satisfy the optimality conditions for detection operators in section 4 . Thus, detection operators (24) are optimal.

6 CONCLUSION

For the general problem of minimum-error discrimination, we give a standard SDP method based on the theory of SDP. Then we derive a set of optimality conditions for detection operators. Furthermore, we can directly obtain the optimal value of the correct detection probability and the optimal detection operators numerically by using the software package $SeDuMi1.05$ in $Matlab7.0$. Simulation results indicate that the standard SDP method is very efficient. The method has several advantages. Firstly, it is simple to implement. Secondly, it converges faster and needs fewer iteration numbers than iteration algorithm. Finally, this method may become a useful tool in quantum state discrimination theory in general.

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